MOHAMMED SAYED ABDELMONEIM MAHMOUD

ELECTRICAL ENGINEER | R&D | POWER ELECTRONICS | RENEWABLE ENERGY OPTIMIZATION

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Electrical Engineer with a Master's degree in Renewable Energy Optimization, offering practical expertise in power electronics, embedded systems, and control design. Conducted a funded master's thesis on photovoltaic system sizing for combined electrical and thermal loads in remote areas. Demonstrated proficiency in circuit design, microcontroller programming, and laboratory prototyping. Currently leading the development of an industrial-grade Variable Frequency Drive (VFD). Published in peer-reviewed venues on energy optimization and load profiling. Committed to driving innovation in sustainable energy, microgrid control, and digital energy systems.

EDUCATION

MSc, Assiut University
Sep 2022 – Aug 2025

- Designed and simulated hybrid energy systems integrating photovoltaic modules, battery storage, and thermal loads. Developed control algorithms and conducted load estimation for rural energy applications. Contributed to a funded research project, resulting in two published papers and one under review, supporting studies in energy access.

BSc, Assiut University
Sep 2016 – Jul 2021

- Graduation Project: Building Management System Based on Mobile Application — Awarded Excellent Grade with Full Marks

CERTIFICATES AND COMPETITIONS

- Young Innovators, African Entrepreneurship and Innovation Forum
- Best Newcomers Award, EVER Egypt 2020
- First Place Winner, Abu Dhabi University Innovation Competition

ORGANIZATIONS

- Smart Grid Lab, Assiut University

PROFESSIONAL EXPERIENCE

Power Electronics Engineer
Nov 2023 – Present

- Designed custom PCBs and converter circuits for clients in both academia and industry
- Provided simulation services, system testing, and control tuning for hardware development teams

Team Member, MSc Research Project – Stand-Alone Green Hydrogen Energy System
Jul 2022 – Aug 2025

- Contributed to a funded research project integrating photovoltaic systems, fuel cells, and electrolyzers
- Designed power electronic interfaces, DC/AC converters, and control systems
- Supported laboratory setup, load profiling, and hydrogen burner development
- Published two research papers; one additional manuscript under review

Project Leader, VFD Innovation Project – Industrial Development
Apr 2023 – Jul 2025

- Leading an R&D initiative to develop a Variable Frequency Drive (VFD) system for industrial applications
- Responsible for hardware design, control system development, and MATLAB/Simulink simulation
- Overseeing microcontroller programming, system integration, and prototype testing
- Coordinating cross-functional teams for product development and deployment

Smart Building & EV Projects

- Developed an IoT-based Building Management System prototype using Simulink and mobile control
- Supported an EV rally competition team with power systems integration and electrical wiring
- Contributed to renewable energy modeling and PV system control tuning projects

TECHNICAL SKILLS

Power Electronics

- Design and analysis of DC/AC converters, electrolyzer circuits, motor control systems, and protection/sensing circuitry.

Hardware Design & Validation

- Proficient in schematic capture, PCB layout, and validation using Altium Designer, Eagle, Proteus, and EasyEDA.

Energy Management

- Expertise in load profiling, energy audits, and rural demand-side management for optimized power distribution.

Model-Based Development

- Skilled in MATLAB/Simulink modeling and simulation for variable frequency drives (VFDs) and renewable energy systems.

Laboratory & Test Equipment

- Hands-on experience with oscilloscopes, power analyzers, Taraz devices, signal generators, and data acquisition systems.

Hydrogen Energy Systems

- Knowledgeable in electrolyzer design, HHO dry cell fabrication, and hydrogen-based burner performance analysis.

LANGUAGES

- **Arabic** (Native/Fluent)
- **English** (Professional Proficiency)

REFERENCES

- **All Available Upon Request.**